

Multiple Choice (10 marks)

1. Why are Cepheid stars relevant for astronomers?
 - (a) To measure interstellar mass.
 - (b) To measure galactic distances.
 - (c) To measure galactic energy-density.
 - (d) To measure interstellar density.
2. What is the Cassini division of Saturn's rings?
 - (a) a dark ring visible from Earth composed of dark dusty particles
 - (b) the most opaque ring of Saturn made of highly reflective ice particles
 - (c) the widest ring of Saturn located between two large ring gaps
 - (d) a large gap visible from Earth produced by an orbital resonance with the moon Mimas
3. The sky is blue because
 - (a) the Sun mainly emits blue light.
 - (b) the atmosphere absorbs mostly blue light.
 - (c) molecules scatter red light more effectively than blue light.
 - (d) molecules scatter blue light more effectively than red light.
4. The astronomical unit parsec (pc) plays a crucial role in astronomy. One parsec is equal to about 3.26 light-years. How is one parsec defined in astronomy?
 - (a) Distance at which one astronomical unit measures one arcsecond from Earth.
 - (b) Orbital distance of the solar system around the center of the Milky Way in one year.
 - (c) Effective distance of the solar wind (i.e. the radius of the heliosphere).
 - (d) Historical distance to the brightest star Sirius.
5. Approximately how old is the surface of Venus?
 - (a) 750 million years.
 - (b) 2 billion years.
 - (c) 3 billion years.
 - (d) 4.5 billion years.

True / False (10 marks)

Answer True or False

6. True or False: Asteroids exhibit substantial motion relative to the background stars, making them distinguishable in sky surveys.
7. True or False: Io is the closest of Jupiter's four largest moons, with the smallest distance to the planet.
8. True or False: On a planet with half the gravity of Earth, a sand bag with a mass of 5 kg would weigh 25 N.
9. True or False: Fresh impact craters invariably possess central peaks, which are a characteristic of all such formations.
10. True or False: Dione is one of the Galilean moons that orbit Jupiter, alongside Ganymede, Callisto, and Europa.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that contribute to Pluto's lengthy orbital period around the Sun, including its distance from the Sun and the implications for its classification as a dwarf planet.
12. Discuss the role of asteroids and comets in the context of the Solar Nebular Theory, focusing on their origins as leftover planetesimals and their significance in planetary formation.

End of Paper